

Harvard Problem¹

Solve for x where...

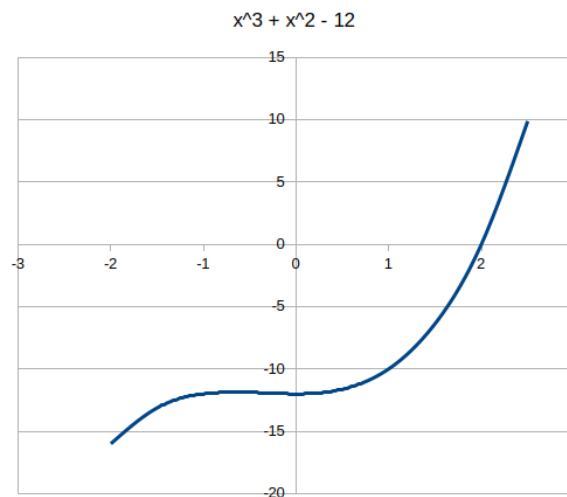
$$x^3 + x^2 = 12$$

*

Solution

We have a cubic equation which can be expressed $x^3 + x^2 - 12 = 0$. There is of course a formula for the roots of such cubics, but it is quite a handful and not many people would memorise it (unlike the well-known quadratic formula).

Examining the graph of this function shows us that there is one real solution, which we can readily see is where $x = 2$. Since the cubic should have three roots we can surmise there are two non-real solutions.



Now, it should be possible to express the equation above in the form:

$$(x - p)(x - q)(x - r) = 0$$

where p , q and r are the roots of the cubic. Since we know one of these values equals 2 (let's say r), we have:

$$(x - p)(x - q)(x - 2) = 0 \quad \equiv \quad x^3 + x^2 - 12$$

or

$$(x - 2)(x^2 - (p+q)x + pq) = 0$$

so

$$\begin{aligned} x^3 - (p+q)x^2 + (pq)x \\ - 2x^2 + 2(p+q)x - 2pq = 0 \end{aligned}$$

Therefore, we can see that the constant $-2pq$ must equal -12 ; that is $q = 6/p$.

Furthermore, since the coefficient of the x term needs be zero:

$$pq + 2p + 2q = 0$$

1 From internet website

so

$$2p^2 + 6p + 12 = 0$$

and, applying the quadratic formula we get:

$$p = \frac{-6 \pm \sqrt{36 - 96}}{4} = \frac{-3 \pm \frac{\sqrt{15}}{2}i}{2} *$$

Alternatively, taking the coefficient of the x^2 term:

$$-(p + q + 2) = 1$$

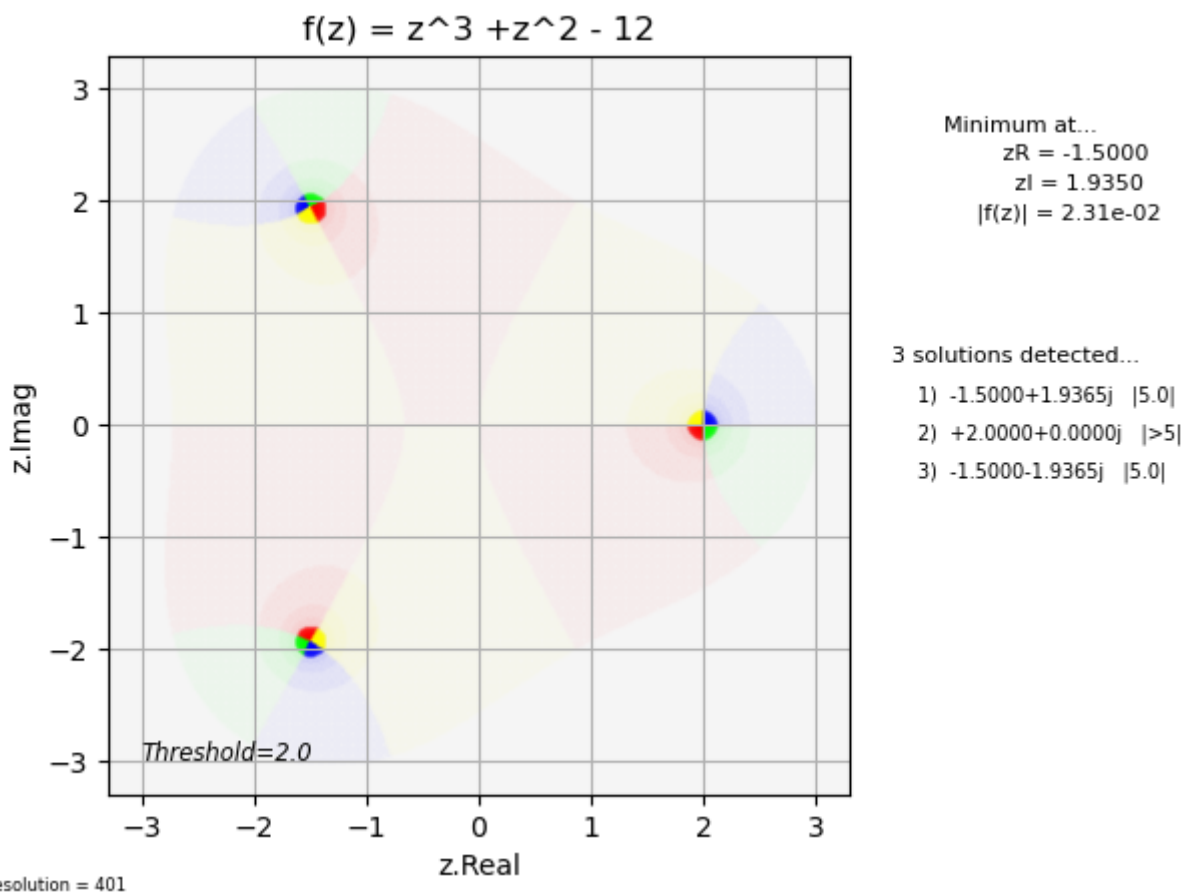
so

$$p^2 + 3p + 6 = 0$$

thus

$$p = \frac{-3 \pm \sqrt{9 - 24}}{2} = \frac{-3 \pm \frac{\sqrt{15}}{2}i}{2} *$$

As evidenced by fRootMaster...



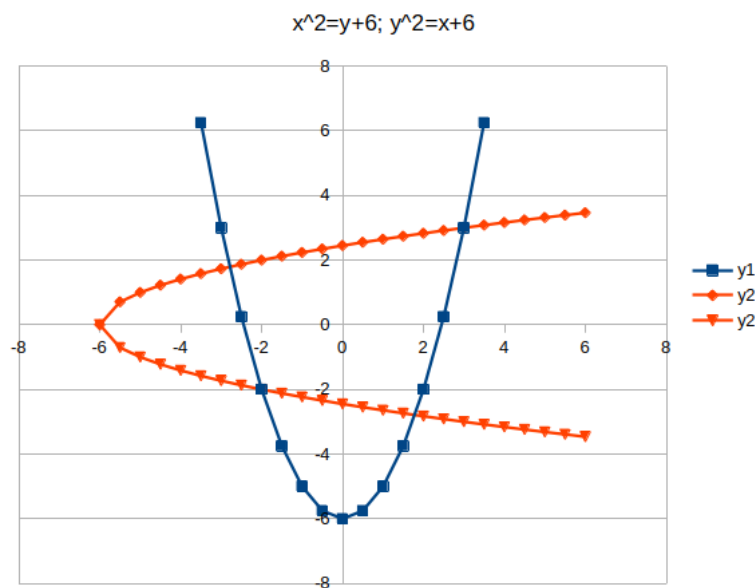
AMS Problem²Solve for x, y given...

$$x^2 = y + 6; \quad y^2 = x + 6$$

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Solution

Sketching these relationships shows...

We can readily see that there are solutions where $x, y = 3, 3$ and $-2, -2$.

From the assertion:

$$x^2 = y + 6; \quad y^2 = x + 6$$

we eliminate y thus:

$$x^2 = \sqrt{x+6} + 6$$

or

$$\sqrt{x+6} = x^2 - 6$$

so

$$x+6 = x^4 - 12x^2 + 36$$

hence

$$0 = x^4 - 12x^2 - x + 30 \quad \dots(i)$$

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Now we have a fourth order polynomial for which we know two solutions. So we can say...

$$(x-3)(x+2)(x-u)(x-v) = 0$$

where u and v are roots...

$$(x^2 - x - 6)(x^2 - (u+v)x + uv) = 0 \quad \dots(ii)$$

2 From text *Algebra Made Simple*

From this we see that for the constant terms in (i) and (ii) to equate, $-6uv$ must equal 30, or...

$$uv = -5 \rightarrow v = -5/u$$

And for the third order terms' coefficients to agree (zero in (i))...

$$u + v - 1 = 0$$

so

$$u - 5/u - 1 = 0$$

or

$$u^2 - u - 5 = 0$$

solving for which...

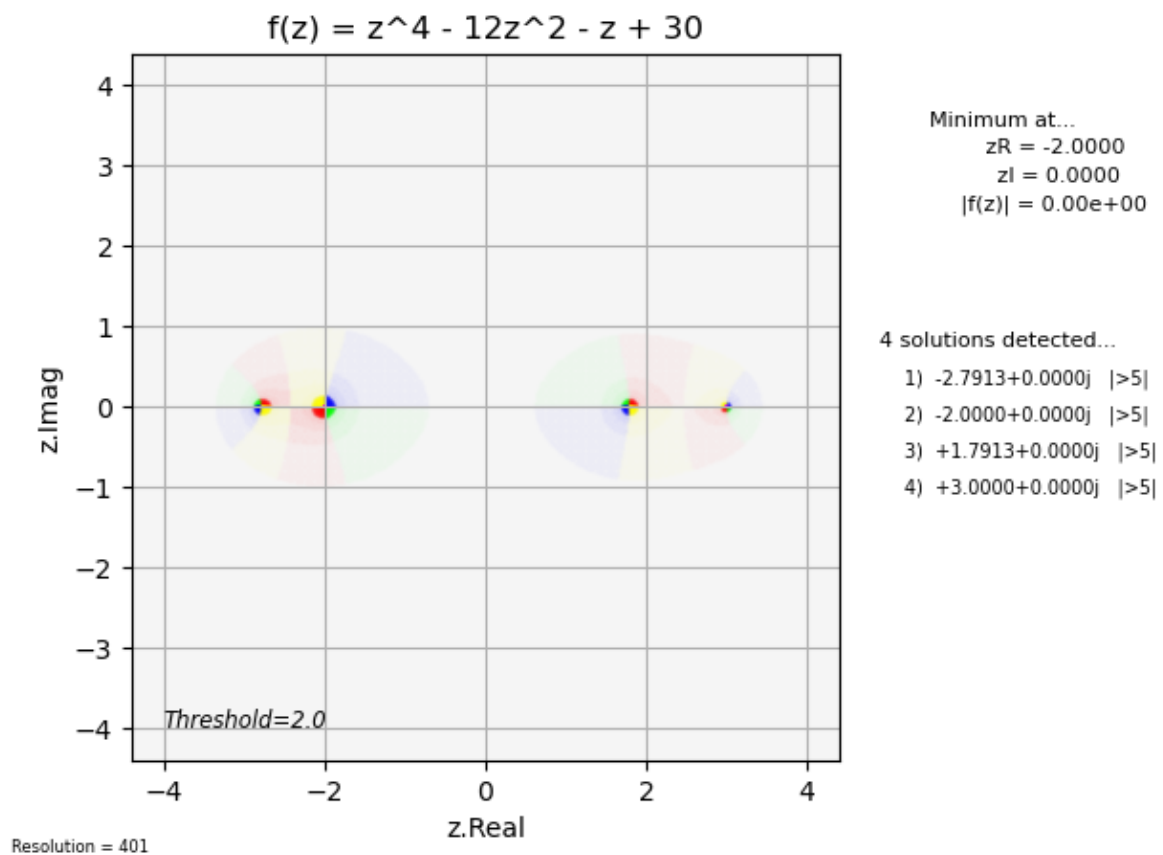
$$u = \frac{1 \pm \sqrt{1+20}}{2} \quad \sim 2.7913, -1.7913$$

and therefore

$$v = \frac{-10}{1 \pm \sqrt{21}}$$

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And from fRootMaster...



Exponential Problem³

Find a where...

$$2^a + 2^{-a} = 1$$

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Solution

Solving for...

$$2^a + 2^{-a} - 1 = 0$$

let $x = 2^a$, then –

$$x + 1/x - 1 = 0$$

so

$$x^2 - x + 1 = 0$$

therefore –

$$x = \frac{1 \pm \sqrt{1-4}}{2}$$

$$x = \frac{1}{2} \pm \frac{\sqrt{3}i}{2}$$

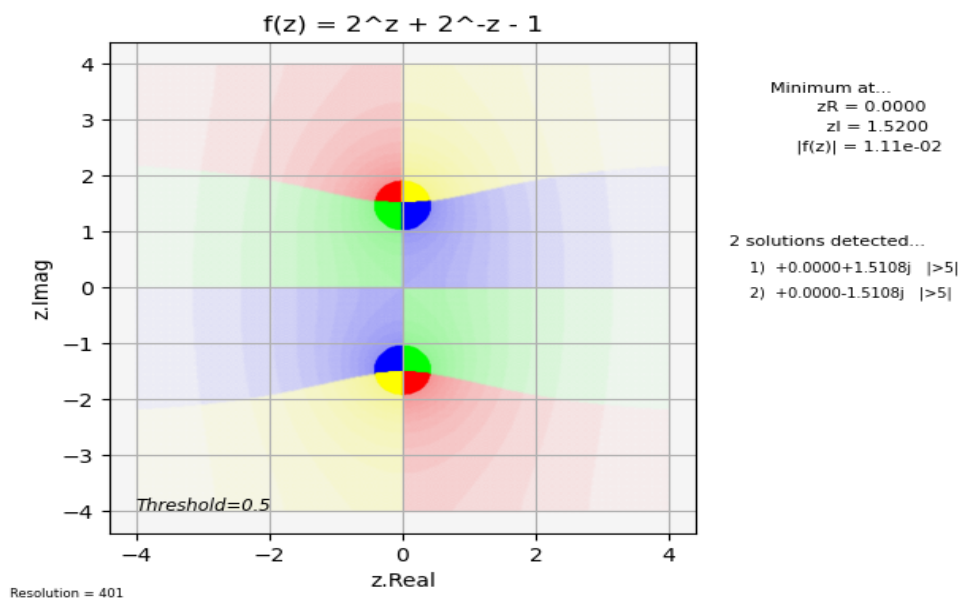
Now

$$a = \ln x / \ln 2$$

$$a \approx \pm 1.047197i / 0.693147 \approx \pm 1.5108i$$

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Confirmed by fRootMaster...



Notes

It may seem remarkable that the solutions work out to equal purely imaginary numbers, i.e. having no real components. However, if we consider the general case for $2^a + 2^{-a} = b$ where $|b| < 2$, we see that the quadratic above becomes:

$$x^2 - bx + 1 = 0$$

and so

$$x = \frac{b \pm \sqrt{b^2 - 4}}{2}$$

$$x = \frac{b}{2} \pm \frac{\sqrt{4 - b^2}i}{2}$$

Now, to calculate the logarithm of x , we need to transform this expression to polar form $x = r \cdot \text{cis}(\theta)$, where

$$r^2 = \frac{b^2}{4} + \frac{4 - b^2}{4}, \text{ that is } r = 1$$

and

$$\theta = \text{atan} \frac{\sqrt{4 - b^2}}{b}$$

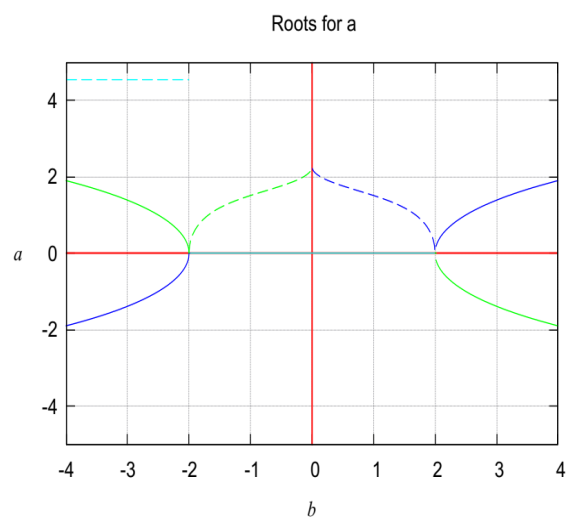
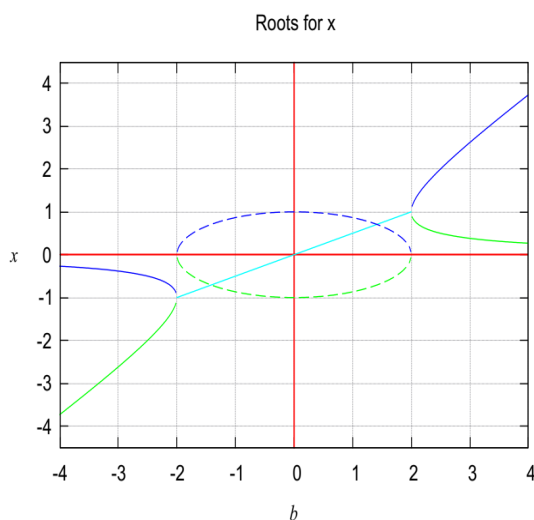
As previously demonstrated, the logarithm of a complex number in this format is equal to: $\ln r + i\theta$. Since the log of 1 is zero, this yields the zero real value. The imaginary value lies in the range π to $-\pi$.

For $b > 2$ the quadratic root becomes: $x = \frac{b \pm \sqrt{b^2 - 4}}{2}$ which is real and positive, thus the solutions for a are real. When $b < -2$ the quadratic root is real and negative, so a solutions become complex values.

Clearly, when $b = \pm 2$, $a = 0$.

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Gnuplots...⁴



KDT May 2026

⁴ Dashed lines represent imag. values